

padding oracle

A padding oracle attack is a side channel attack

An attacker submits modified cipher text to a server. The server decrypts and reveals through error message or timings - ~~total~~ whether padding is valid or not.

This 'oracle' leaks one bit of info per query. By systematically flipping bits an observer, valid / invalid padding and attacker can reverse engineer the plaintext without knowing the key.

Caused by

Enc without authentication, the system decrypts first then checks padding in to avoid internal state